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Qualification Pack

Road Construction Engineer

QP Code: ASP/CON/Q0201

Version: 1.0

NSQF Level: 5.5

Additional Skill Acquisition Programme Kerala || Additional Skill Acquisition Programme Kerala, 3rd
Floor, Trans Towers, Vazhuthacaud
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Qualification Pack

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ASP/CON/Q0201: Road Construction Engineer

Brief Job Description

Road construction Engineers are professionals responsible for the overseeing, costing, estimating, planning as well as implementation of construction projects of roads that connect cities, towns, and states. They play a pivotal role in overseeing all aspects of road construction, from the initial planning and design stages to construction and project completion. Their work ensures that roads are built safely, efficiently, and in compliance with the regulations.

Personal Attributes

Attention to Detail ,Problem-Solving Skills ,Project Management Skills, Knowledge of Materials & Construction Methods, Familiarity with Regulations

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ASP/CON/N0201: Overview on Road Design and Construction](#)
2. [ASP/CON/N0202: Overview of Construction Tools, Machinery and Practices](#)
3. [ASP/CON/N0203: Construction of Embankment and Granular Layers](#)
4. [ASP/CON/N0204: Construction of Pavement Upper Layers and Safety Measures](#)
5. [ASP/CON/N0205: Geotechnical Investigations, Construction of Bridges and Cross Drainage Structures](#)
6. [ASP/CON/N0206: Health and Safety in Road construction](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Construction
Sub-Sector	REAL ESTATE AND INFRASTRUCTURE CONSTRUCTION
Occupation	Highway Engineer
Country	India

Qualification Pack

NSQF Level	5.5
Credits	21
Aligned to NCO/ISCO/ISIC Code	2142.0600
Minimum Educational Qualification & Experience	B.E./B.Tech (CIVIL ENGINEERING) with NA of experience OR Completed 3 year diploma after 10th (IN CIVIL ENGINEERING) with 5 Years of experience OR Completed 2nd year diploma after 12th (in Civil Engineering) with 3 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	22 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	1.0
Reference code on NQR	QG-5.5-CO-03517-2025-V1-ASAP
NQR Version	1

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ASP/CON/N0201: Overview on Road Design and Construction

Description

Road design and construction involve planning, engineering, and building roads to ensure safe, efficient, and durable transportation infrastructure.

Scope

The scope covers the following :

- The scope of this module covers the fundamental principles of road design and construction, including geometric design, material selection, pavement structure, drainage, and safety considerations. It also explores modern engineering techniques, sustainability practices, and regulatory standards for efficient road infrastructure development.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Assess the basic understanding of road construction principles
- PC2.** Evaluate proficiency in conducting topographic surveys using GPS, Total Station, and Auto Level, including accurate site setting out.
- PC3.** Assess an overall knowledge of DPR Preparation
- PC4.** Assess the knowledge in interpretation of Drawings in the DPR
- PC5.** Evaluate the proficiency in the Geometric and Pavement Design
- PC6.** Assess the knowledge about Labour Laws, Project Management- CPM, PERT , MS Project, Primavera, Building Information Modeling (BIM)

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	69	-	1
PC1. Assess the basic understanding of road construction principles	5	-	-	-
PC2. Evaluate proficiency in conducting topographic surveys using GPS, Total Station, and Auto Level, including accurate site setting out.	5	25	-	-
PC3. Assess an overall knowledge of DPR Preparation	5	10	-	1
PC4. Assess the knowledge in interpretation of Drawings in the DPR	5	15	-	-
PC5. Evaluate the proficiency in the Geometric and Pavement Design	5	10	-	-
PC6. Assess the knowledge about Labour Laws, Project Management- CPM, PERT , MS Project, Primavera, Building Information Modeling (BIM)	5	9	-	-
NOS Total	30	69	-	1

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/CON/N0201
NOS Name	Overview on Road Design and Construction
Sector	Construction
Sub-Sector	
Occupation	Highway Engineer
NSQF Level	5.5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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ASP/CON/N0202: Overview of Construction Tools, Machinery and Practices

Description

This NOS provides a comprehensive understanding of road construction processes, including the selection, operation, and maintenance of construction tools and machinery. It equips learners with the skills needed to manage preparatory works, utility services, and material handling efficiently. The NOS emphasizes best practices in construction techniques, sustainability, and safety compliance. It also covers quality management through field laboratory operations, testing, and inspections to ensure adherence to project specifications. By integrating practical training with industry standards, this NOS prepares professionals to contribute effectively to road construction projects with a focus on quality, efficiency, and environmental responsibility.

Scope

The scope covers the following :

- This NOS covers the fundamental aspects of road construction engineering, focusing on the essential tools, machinery, and best practices required for efficient project execution. It includes preparatory works, construction techniques, quality management, utility management, and material selection. The scope also extends to ensuring safety, sustainability, and compliance with industry standards through the effective use of construction equipment and quality control measures.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Evaluate the understanding of Quality Management Plan -Setting up of Field laboratory, Quality Assurance & Quality Control
- PC2.** Assess the understanding of Site Clearance-Earthwork, Erosion Control and Drainage- Quality Control
- PC3.** Assess the knowledge of Construction Tools and Machinery- Types, Applications, Suitability and Selection. Assess the knowledge on Full Depth Reclamation (FDR), precast pavements, white topping and surface dressing
- PC4.** Evaluate the knowledge of Material Selection and Management, utility management

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	78	-	2
PC1. Evaluate the understanding of Quality Management Plan -Setting up of Field laboratory, Quality Assurance & Quality Control	3	10	-	-
PC2. Assess the understanding of Site Clearance- Earthwork, Erosion Control and Drainage- Quality Control	5	23	-	1
PC3. Assess the knowledge of Construction Tools and Machinery- Types, Applications, Suitability and Selection. Assess the knowledge on Full Depth Reclamation (FDR), precast pavements, white topping and surface dressing	6	20	-	1
PC4. Evaluate the knowledge of Material Selection and Management, utility management	6	25	-	-
NOS Total	20	78	-	2

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/CON/N0202
NOS Name	Overview of Construction Tools, Machinery and Practices
Sector	Construction
Sub-Sector	
Occupation	Highway Engineer
NSQF Level	5.5
Credits	4
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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ASP/CON/N0203: Construction of Embankment and Granular Layers

Description

This NOS provides a comprehensive understanding of the methods and techniques required for constructing and managing embankments, subgrades, and granular layers in road construction. It covers material specifications, grading, compaction, and stabilization techniques to ensure durability and performance. The NOS emphasizes quality assurance through laboratory and field testing of soils and aggregates while integrating environmental sustainability measures. Learners will gain hands-on experience with surveying, excavation, grading, and material testing equipment. Additionally, this NOS includes fundamental design principles for earth retaining works, ensuring stability and structural integrity. Traffic management and worker safety measures are also highlighted to minimize risks during construction.

Scope

The scope covers the following :

- This NOS covers the essential principles and practices involved in the construction of subgrade, embankments, granular layers, and earth retaining structures in road projects. It encompasses material selection, compaction techniques, quality control measures, environmental considerations, and traffic management strategies. The scope also includes the use of appropriate construction equipment, adherence to safety protocols, and the implementation of best practices to ensure compliance with industry standards and project specifications.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Check the understanding of Embankment and Subgrade - Design Criteria, Construction and Quality Control
- PC2.** Assess the knowledge about Field and Laboratory Testing of Soil, Aggregates and Other Additives/ Stabilizing Agents
- PC3.** Assess the understanding of Earth Retaining Works- Types, Basic Design Principles and Construction Practices
- PC4.** Assess the knowledge about Sub-Bases, Bases (Non-Bituminous) and Shoulder - Types, Material Specifications and Construction Practices

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	83	-	2
PC1. Check the understanding of Embankment and Subgrade - Design Criteria, Construction and Quality Control	3	16	-	-
PC2. Assess the knowledge about Field and Laboratory Testing of Soil, Aggregates and Other Additives/ Stabilizing Agents	5	21	-	1
PC3. Assess the understanding of Earth Retaining Works- Types, Basic Design Principles and Construction Practices	2	23	-	1
PC4. Assess the knowledge about Sub-Bases, Bases (Non-Bituminous) and Shoulder - Types, Material Specifications and Construction Practices	5	23	-	-
NOS Total	15	83	-	2

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/CON/N0203
NOS Name	Construction of Embankment and Granular Layers
Sector	Construction
Sub-Sector	
Occupation	Highway Engineer
NSQF Level	5.5
Credits	4
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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ASP/CON/N0204: Construction of Pavement Upper Layers and Safety Measures

Description

This NOS provides a detailed understanding of the principles and techniques involved in constructing and maintaining the upper layers of pavements. It includes the specifications and requirements for bituminous and concrete pavements, covering aspects such as design mix, reinforcement, and thickness standards. The NOS emphasizes the correct use of equipment, such as asphalt pavers, rollers, and concrete mixers, to ensure the efficient execution of pavement construction. Safety measures are integrated throughout the process to protect workers and the public, while traffic management strategies ensure smooth operations. Learners will gain hands-on experience in layer compaction, pavement marking, and installation of road safety elements to enhance the overall quality and functionality of road infrastructure.

Scope

The scope covers the following :

- This NOS covers the construction and maintenance of pavement upper layers, including bituminous and concrete surfaces. It includes best practices for material selection, layer installation, compaction, and quality control to ensure pavement durability and strength. The scope also encompasses the proper use of construction equipment, adherence to industry standards, and implementation of safety measures for workers and road users. Additionally, the NOS highlights the importance of traffic management and road safety through the correct installation and maintenance of traffic signs, markings, and other road appurtenances.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** - Assess the knowledge about Bituminous Bases and Surface Courses - Specifications and Construction Practices. Evaluate Local/New/Alternate/Modified Construction Materials and Technologies - Use of Waste Plastic, Full Depth Reclamation (FDR), Reclaimed Asphalt Pavement (RAP) etc; Bitumen testing- and Test on mixes- Marshall stability, etc.
- PC2.** Evaluate the knowledge of Concrete Pavement - Specification and Construction Practices
- PC3.** Understanding of Traffic Signs, Markings and Other Road Appurtenances
- PC4.** Understanding of Construction Equipments and Machinery for base and surface courses

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	77	-	3
PC1. - Assess the knowledge about Bituminous Bases and Surface Courses - Specifications and Construction Practices. Evaluate Local/New/Alternate/Modified Construction Materials and Technologies - Use of Waste Plastic, Full Depth Reclamation (FDR), Reclaimed Asphalt Pavement (RAP) etc; Bitumen testing- and Test on mixes- Marshall stability, etc.	5	15	-	1
PC2. Evaluate the knowledge of Concrete Pavement - Specification and Construction Practices	5	22	-	1
PC3. Understanding of Traffic Signs, Markings and Other Road Appurtenances	5	20	-	1
PC4. Understanding of Construction Equipments and Machinery for base and surface courses	5	20	-	-
NOS Total	20	77	-	3

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/CON/N0204
NOS Name	Construction of Pavement Upper Layers and Safety Measures
Sector	Construction
Sub-Sector	
Occupation	Highway Engineer
NSQF Level	5.5
Credits	4
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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ASP/CON/N0205: Geotechnical Investigations, Construction of Bridges and Cross Drainage Structures

Description

This NOS provides a comprehensive understanding of geotechnical investigations and their role in infrastructure development. Learners will gain knowledge of soil and rock sampling techniques, laboratory and in-situ testing methods, and geotechnical analysis to inform construction decisions. The module also covers the construction processes for bridges and cross drainage structures, including excavation, piling, concrete placement, reinforcement installation, and structural erection. Emphasis is placed on ensuring compliance with industry standards, contractual obligations, and safety regulations. Practical training will include hands-on experience in geotechnical assessments, foundation design, and the construction of bridge components such as piers, abutments, girders, and decks. The NOS also includes project completion procedures, final inspections, and site handover best practices.

Scope

The scope covers the following :

- This NOS covers geotechnical investigations and the construction of bridges and cross drainage structures.
- It includes the study of soil and rock properties, groundwater conditions, and other geotechnical parameters necessary for designing stable foundations, slopes, and retaining structures.
- The scope also encompasses the execution of construction activities for substructures and superstructures, ensuring compliance with regulatory standards and quality requirements. It highlights the importance of safety, proper documentation, and final inspections before project handover.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** • Assess the knowledge of Surface and Sub-Surface Geotechnical Investigations- Principles
• and Practices
- PC2.** Access knowledge of Foundations - Types, Material Specifications, Testing and Construction Practices
- PC3.** Evaluate the understanding of Construction of Substructures and Super Structures of Bridges and Cross Drainage Structures, Formworks
- PC4.** Evaluate Steel Reinforcement, Prestressing, Bearings, Expansion Joints, Appurtenances, Repair and Retrofitting of Structures
- PC5.** Access the knowledge of the Completion of Work and Site Handing Over

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	83	-	2
PC1. • Assess the knowledge of Surface and Sub-Surface Geotechnical Investigations- Principles • and Practices	5	13	-	1
PC2. Access knowledge of Foundations - Types, Material Specifications, Testing and Construction Practices	2	15	-	1
PC3. Evaluate the understanding of Construction of Substructures and Super Structures of Bridges and Cross Drainage Structures, Formworks	4	25	-	-
PC4. Evaluate Steel Reinforcement, Prestressing, Bearings, Expansion Joints, Appurtenances, Repair and Retrofitting of Structures	2	25	-	-
PC5. Access the knowledge of the Completion of Work and Site Handing Over	2	5	-	-
NOS Total	15	83	-	2

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/CON/N0205
NOS Name	Geotechnical Investigations, Construction of Bridges and Cross Drainage Structures
Sector	Construction
Sub-Sector	
Occupation	Highway Engineer
NSQF Level	5.5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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ASP/CON/N0206: Health and Safety in Road construction

Description

This NOS provides learners with comprehensive knowledge and practical applications related to health and safety in road construction, emphasizing regulatory compliance, hazard assessment, and the use of personal protective equipment (PPE). It also includes employability skills essential for workforce readiness, such as time management, problem-solving, communication, and digital literacy. Learners will gain insights into financial management, professional ethics, business planning, and sustainable work practices. The module prepares individuals to navigate career opportunities, conduct safe financial transactions, and engage in effective teamwork while maintaining workplace hygiene and etiquette.

Scope

The scope covers the following :

- This NOS covers essential health and safety practices in road construction and the employability skills required for career readiness. It includes identifying and mitigating worksite hazards, adhering to industry safety regulations, and implementing emergency response protocols. Additionally, it focuses on developing professional skills such as communication, teamwork, financial literacy, digital proficiency, and entrepreneurship, ensuring learners are equipped for sustainable career growth in the road construction industry.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Evaluate the knowledge of Safety Practices in Construction

PC2. Evaluate the knowledge of Precautionary Measures for Safeguarding the Environment, Traffic Management Measures During Construction

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	40	-	-
PC1. Evaluate the knowledge of Safety Practices in Construction	5	20	-	-
PC2. Evaluate the knowledge of Precautionary Measures for Safeguarding the Environment, Traffic Management Measures During Construction	5	20	-	-
NOS Total	10	40	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/CON/N0206
NOS Name	Health and Safety in Road construction
Sector	Construction
Sub-Sector	
Occupation	Highway Engineer
NSQF Level	5.5
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment Components & Weightage

ASAP Kerala employs a structured assessment framework that includes Attendance, Assignments, Internal Assessments, and Final Assessments. Assessments align with the National Skills Qualifications Framework (NSQF) and follow a Continuous and Comprehensive Evaluation (CCE) approach.

Theory(40% Weightage) Practical(60% weightage)

Assessment Type Max Marks Assessment Type Max Marks

Assignment 1 30 Lab Assignment 30

Internal Assessment 1 60 Internal Assignment 60

Assignment 2 30 Internship/Lab Assessment 100

Internal Assessment 2 80 Final Project+Viva 100

Final Assessment 100 Attendance 10

Qualification Pack

Total 300 Total 300

Attendance Criteria

Skill based sessions and training are delivered through lectures, discussions, demonstrations, and experiments ensuring student participation during daily learning activities. So those who miss such sessions will not be able to compensate for the loss. Regular student participation in daily classroom activities plays a significant role in student's success. Students whose attendance fall below 70% will not be eligible for final assessment and course completion certificate.

Attendance % Marks

Above 90% 10

86-90% 9

80-85% 8

70-80% 7

Below 70% 0

Internal/Continuous Assessments & Assignments

Assignments (35 marks each): Two assignments, after 25% and 75% of syllabus completion. Evaluated and recorded by trainers.

Internal Assessments (50 marks each): Conducted after 50% and 100% of syllabus completion. Questions include 60% objective and 40% descriptive formats.

Final/End Assessment

Conducted by the Assessment Division of ASAP Kerala, covering:

Practical/Skill Test (at OJT/Skill Development Centers)

Project Presentation & Viva Voce

Qualification Pack

MCQ & Descriptive Written Exam

Assessment Governance & Quality Assurance

Assessment Division of ASAP Kerala oversees the process.

Question Bank prepared by Subject Matter Experts; 20% of questions refreshed annually.

Assessments held at ASAP Skill Parks, Accredited Centers, Colleges.

ID Verification .

Final Score & Certification issued post evaluation by expert assessors.

Final Score & Grading

Score (%) Grade Description

90-100 A+ Excellent

80-89 A Very Good

70-79 B Good

60-69 C Above Average

50-59 P Pass

Below 50 F Fail

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Qualification Pack

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASP/CON/N0201.Overview on Road Design and Construction	30	69	0	1	100	10
ASP/CON/N0202.Overview of Construction Tools, Machinery and Practices	20	78	0	2	100	20
ASP/CON/N0203.Construction of Embankment and Granular Layers	15	83	0	2	100	20
ASP/CON/N0204.Construction of Pavement Upper Layers and Safety Measures	20	77	0	3	100	20
ASP/CON/N0205.Geotechnical Investigations, Construction of Bridges and Cross Drainage Structures	15	83	0	2	100	10
ASP/CON/N0206.Health and Safety in Road construction	10	40	0	0	50	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	130	460	-	10	600	100

Logo
Not
Available



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.